

QUANTUM ALGORITHMS

Mind the Gap

Achieving a super-Grover quantum speedup by jumping to the end

Speaker Name — Institution • Quantum Algorithms Seminar

Setup: combinatorial optimization

Combinatorial optimization

- Classical cost function $H: \{+1, -1\}^n \rightarrow \mathbb{R}$
- Goal: minimize cost, find z^* such that $E^* = H(z^*)$ is minimized

Grover's algorithm

- Solves binary optimization problems in time $O^*(2^{0.5n})$
- Unlikely to yield practical quantum advantage due to
 - Slower quantum clock speeds
 - Error-correction overheads
 - Lack of parallelizability

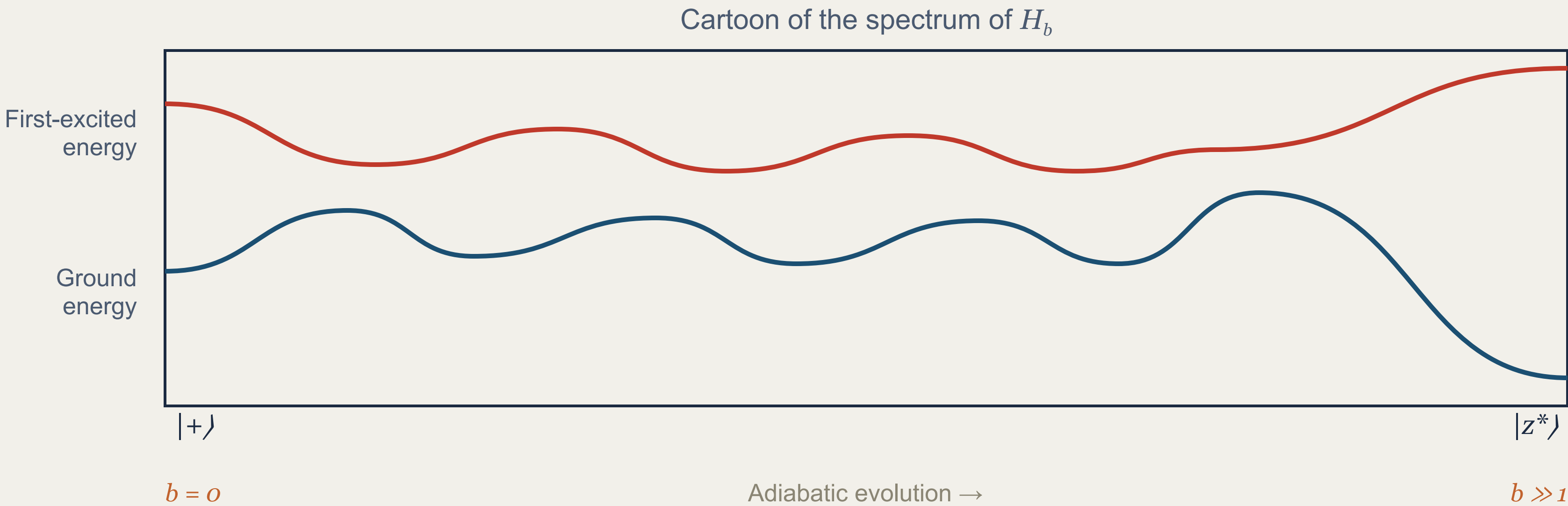
See: [1] Campbell, Khurana, Montanaro (2019) arXiv:1810.05582 · [2] Babbush et al. (2021) arXiv:2011.04149

Warm up: the adiabatic algorithm for combinatorial optimization

- Evolve by Hamiltonian H_b
- Tune from $b = 0$ to $b \gg 0$

$$H_b = -\frac{X}{n} + b \frac{H}{|E^*|} \quad \text{where } X = \sum_{i=1}^n X_i$$

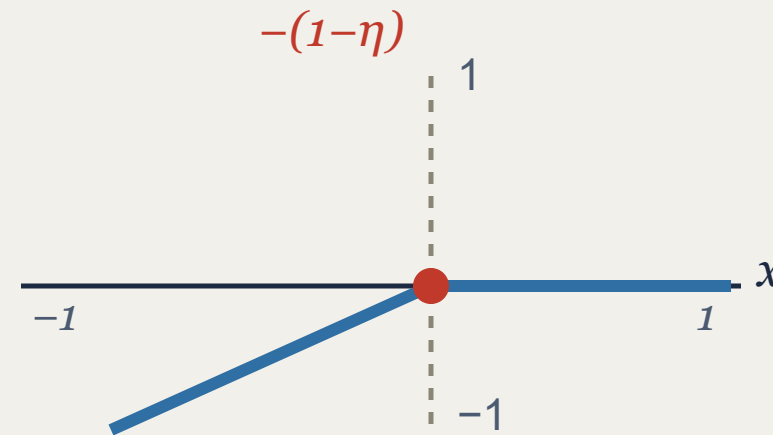
Adiabatic evolution and the spectral gap



Two simple modifications to the adiabatic algorithm

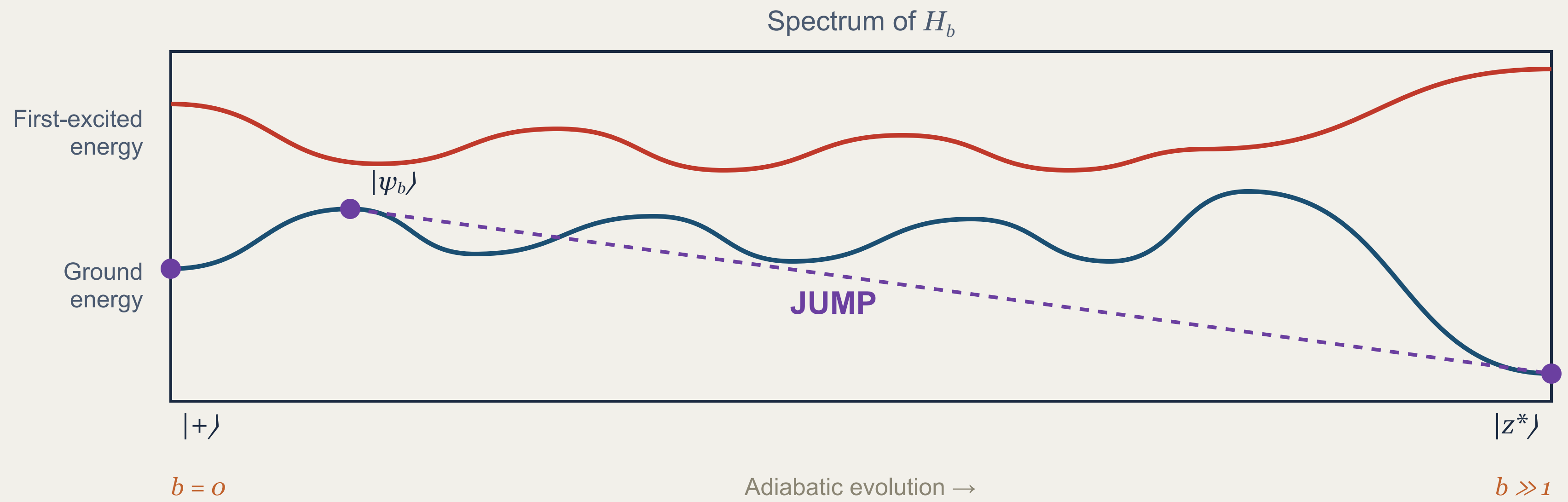
1. **Change the cost function** so almost all assignments have zero cost, while cost structure is preserved in the deepest valleys

$$H_b = -\frac{X}{n} + b g_\eta \left(\frac{H}{|E^*|} \right)$$



2. **Replace the slow final anneal** with a direct jump: measure H_b via phase estimation and amplitude-amplify onto the ground state

Jumping over the small spectral gaps



Full specification of the algorithm

Input: cost function H , optimal value E^* , $\eta < 1$, $b < 1$

Procedure

1. Prepare $|+\rangle$
2. "Jump" to $|\psi_b\rangle$: measure H_b using phase estimation and amplitude-amplify to boost success probability of projecting onto the ground state
3. "Jump" to $|z^*\rangle$: measure in the computational basis to yield $|z\rangle$ and amplitude-amplify the outcome $H(z) = E^*$

Output: $|z^*\rangle$

RECALL

$$H_b = -\frac{X}{n} + b g_\eta \left(\frac{H}{|E^*|} \right)$$

Problem: the algorithm assumes E^* is known

Recall the Hamiltonian used in the jump

$$H_b = -\frac{X}{n} + b g_\eta \left(\frac{H}{|E^*|} \right)$$

- To build H_b , we need the optimal value E^*
- But E^* is **unknown** in real optimization problems — it's the very answer we're trying to find

QUESTION

Can we still run the algorithm without knowing E^ in advance?*

Key insight: search over (θ, φ) , not E^*

Rewrite $b' g_{\eta'}(x / W)$ as $\frac{\varphi}{Q} x + \theta \varphi$

$$\theta = \frac{W(1 - \eta')}{Q} \quad \varphi = \frac{b'Q}{\eta'W}$$

Many different triples (W, η', b') produce the **same Hamiltonian** — so instead of guessing E^* , we only need to search over (θ, φ) .

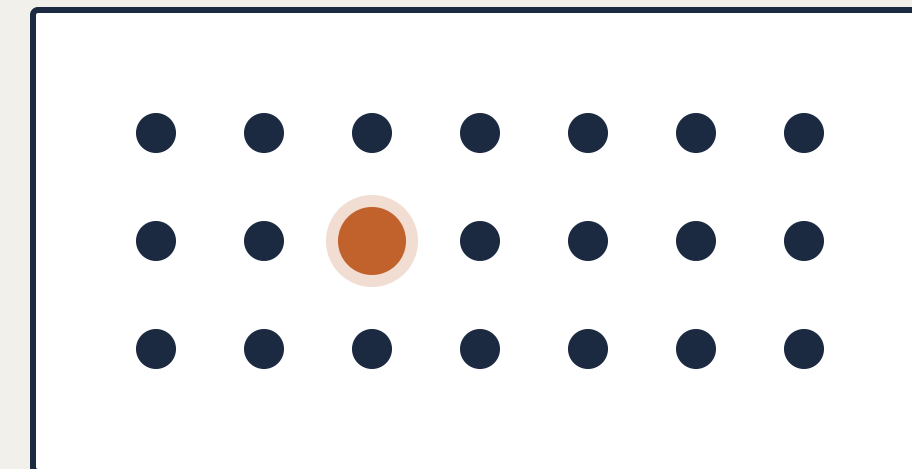
A polynomial-size grid search

The paper proves $\theta \in [0, 1]$ and $\varphi \in [0, Q/q]$

Since $Q/q \leq \text{poly}(n)$, the search region is only **polynomially large**

q, Q — known lower and upper bounds with $q \leq |E^*| \leq Q$ (Prop. 23), so the interval containing E^* is not exponentially large.

- Each grid point defines one candidate Hamiltonian
- Run the short jump at every point — at least one lands close enough to the true parameters



● grid point close enough to the true (θ, φ)

Recovering E^* by binary search

Suppose a grid point prepared the correct state $|\psi_b\rangle$

Instead of checking $H(z) = E^*$, test $H(z) \leq U$ for a guessed threshold U

Only $O(\log(1/\varepsilon))$ iterations are required

Guess $U = 50$ ✓

Guess $U = 25$ ✗

Guess $U = 38$ ✓

Guess $U = 36$ ✗

$E^* = 37$

Putting it together: complexity



Original runtime

$$O^*(2^{(0.5 - c)n})$$

- Grid search adds **poly(n)** overhead
- Binary search adds **$O(\log(1/\epsilon))$** overhead

Therefore, the same runtime holds:

$$O^*(2^{(0.5 - c)n})$$

The asymptotic quantum speedup is preserved.

A short–long–short composition of the two short-path duals

- DPCB23 jumps small-then-large: anchor $|\psi_D\rangle$ sits in the **extended** phase
- Hastings jumps large-then-small: anchor $|\psi_H\rangle$ sits near z^* in the **localized** phase
- Both are gapped ground-state anchors — DPCB23 flags combining mechanisms as the open question

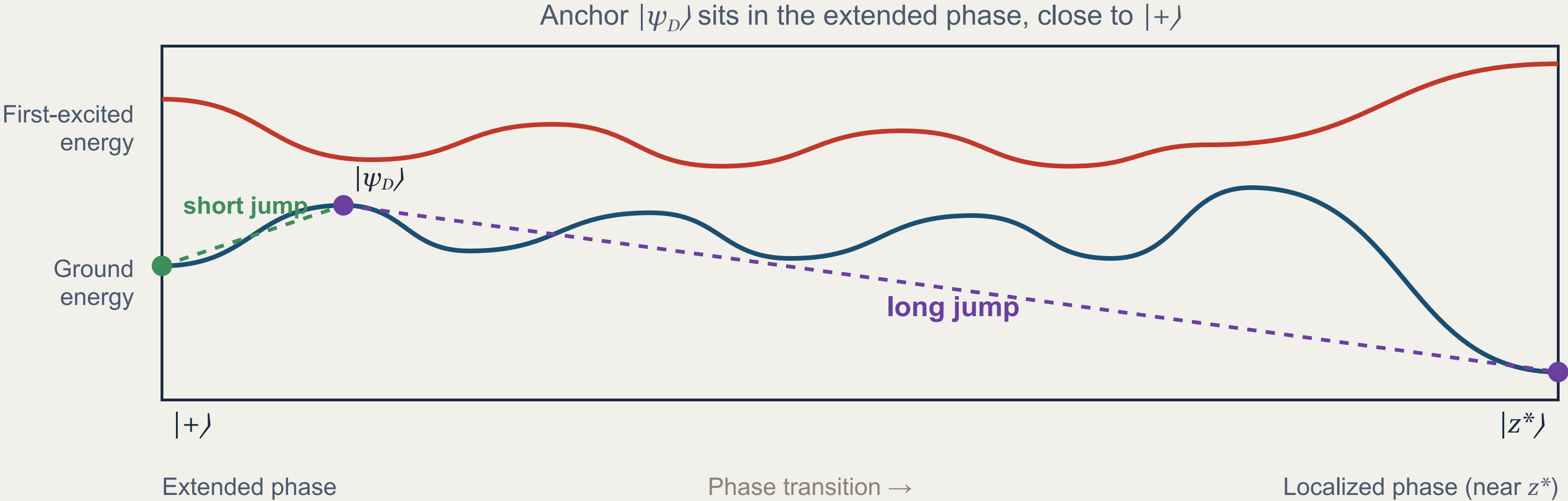
PROPOSAL

Stitch the two duals into a **three-jump scheme**

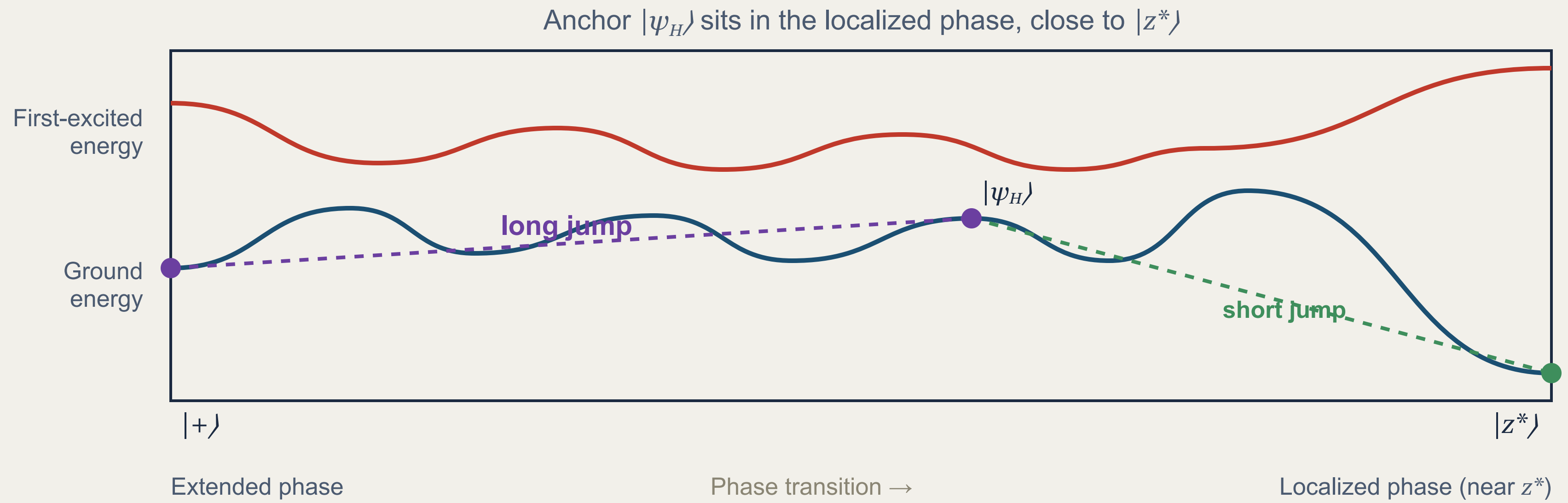
$$|+\rangle \rightarrow |\psi_D\rangle \rightarrow |\psi_H\rangle \rightarrow |z^*\rangle$$

Since both anchors are stoquastic gapped ground states, costs **add** rather than multiply (Prop. 1)

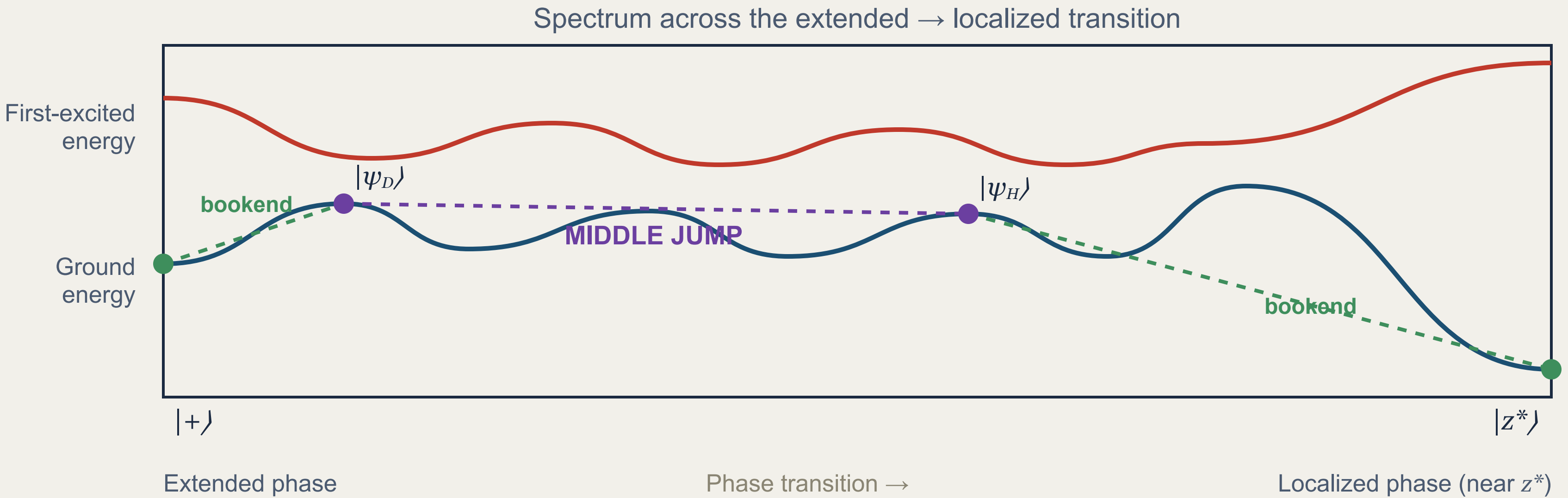
DPCB23: a short jump, then a long jump



Hastings: a long jump, then a short jump



Two cheap bookend jumps, one middle jump



Preliminary numerics: exponents separate near the transition

QUANTITY	EXPONENT / n
direct $\langle z^* \psi_D \rangle$	0.42
middle $\langle \psi_D \psi_H \rangle$	0.20
bookend $\langle z^* \psi_H \rangle$	$\rightarrow 1$
bookend $\langle + \psi_D \rangle$	≈ 1 (flat)
total cost, DPCB23	$2^{0.41n}$
total cost, 3-jump	$2^{0.33n}$

3-spin ensemble, exact diagonalization, $n = 10\text{--}18$, near-critical $s = 0.9$

- Direct-jump exponent reproduces DPCB23's published $2^{0.427n}$ fit — validates the setup
- Reaches **~1.6× advantage** at $n = 18$, with a visibly shallower slope
- Advantage lives only in a near-critical window — deep in either phase the composition is a wash

Preliminary numerical evidence, not yet a proof — see caveats: finite-size transients, both gap assumptions must hold simultaneously.

Outlook: how far can we push the speedup?

- We showed a *provable* super-Grover speedup based on a fundamentally quantum effect
- Potential to produce *super-quadratic* speedups — the kind necessary for practical advantage
- Magnitude of the provable speedup is tiny ($0.5 \rightarrow 0.49983$)
- Numerical results suggest the actual speedup could be larger
- Further investigation of the algorithm could yield even bigger speedups



Thank you

Questions & discussion

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